Formula 1 Drivers and Races: A Statistical Analysis

Savannah Doughty, Jackson Harness, Soraya Remaili

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Dr. David Hitchcock

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I. Background

This dataset, compiled by Ergast Developer API, consists of all relevant information regarding Formula 1 drivers and races from 1950 (the year Formula 1 started) to 2024. Variables including lap times, circuits, qualifiers, pit stops, nationalities, and altitudes provide insight into the popular international racing sport that is Formula 1, or F1. The dataset is packaged into several .csv files, including separate files for drivers, lap times, etc., to make a total of 14 variables. Each .csv file varies in size, with the drivers.csv file containing 862 F1 drivers.

Each 'case' in this dataset represents the result of a particular race for a particular driver. For example, one row could represent the data for Lewis Hamilton's performance in the 2008 Australian Grand Prix or Max Verstappen's performance in the 2023 Dutch Grand Prix. The variables that we will be measuring for each of these cases are as follows. Note that there are two 'nationality' variables, one for the constructor and one for the driver. These will be renamed accordingly in our analysis.

circuitRef - The name of the race circuit
location - The city of the race circuit
altitude - The altitude of the race circuit
constructorRef - The name of the constructor
nationality - The country where the constructor is from
driverRef - The name of the driver
nationality - The country where the driver is from
year - The year of the race
grid - The position the driver started
position - The position the driver ended
points - How many points the driver scored

There are also two additional datasets that detail both the driver standings and the constructor standings at each point in a respective season. Each 'case' in these datasets is also a race result, but is cumulative and can be for either a driver or a constructor. For these datasets, the variables are as follows:

points - How many points the driver/constructor has scored so far
position - The position in the championship

II. Data Set Importance

The Formula 1 season comprises the world's most prestigious motor racing competition. Competition popularity is evergrowing, as between the years of 2021 and 2024, interest in the F1 races grew by 5.7%. This makes F1 racing the fastest growing international competition (Francks, 2025). Economically, Formula 1 depicts a similar success. In 2024, F1 gained a revenue of $3.41 billion - an increase of 5.9% from 2023 (O'Sullivan and Johnson, 2025). The growing attention and profit from F1 racing warrants a better understanding of the factors within this racing, such as an analysis of the races, drivers, and even the locations.

To contribute to the very wide gap of academic analysis of F1 racing, we decided to analyze several factors of F1 racing. By analyzing factors such as location of races, altitude of races, father/son comparisons, strongest "comebacks", and more, we can help the major figures of F1 racing better understand their own field. Knowing the factors that help F1 drivers thrive, F1 executives and sponsors can make economically “safe” decisions for the betterment and growth of Formula 1 competitions.

III. Data Preprocessing and Collection

The data preprocessing stage of the Formula 1 analysis project was carefully structured and began with the importation of multiple .csv files using the read_csv() function from the readr package. These files included data on circuits, constructors, drivers, races, results, and both driver and constructor standings. During import, the data was immediately trimmed using the select() function to retain only the necessary variables for analysis. For example, from the circuits.csv file, only the circuit ID, reference, location, longitude, latitude, and altitude were kept, while the constructors.csv and drivers.csv files were reduced to contain only their respective IDs, references, and nationalities.

Once the data was imported, the next step involved joining and integrating multiple tables to form a comprehensive dataset. The results dataset served as a central hub for this process, being joined sequentially with the races, circuits, drivers, and constructors datasets using the inner_join() function. This enabled a unified structure that combined race context, geographical circuit data, driver and constructor metadata, and race outcomes. Notably, renaming was performed during this stage to prevent ambiguity—driver and constructor nationalities were renamed to driver_nationality and constructor_nationality respectively, as both fields initially shared the same column name.

Further preprocessing was performed to construct a consolidated standings dataset. The driver_standings table was joined with the results_joined dataset to add constructor information, and then subsequently joined with the constructor_standings and races datasets. Variables were renamed accordingly to distinguish between driver-level and constructor-level standings, such as driver_points and const_points, and to preserve clarity when performing further transformations and filtering.

One of the key cleaning operations involved ensuring that all columns used in calculations were in the correct format. For instance, the position column from results_joined—originally read in as a character due to the presence of non-numeric entries—was explicitly converted to numeric using the mutate() function with as.numeric(). This transformation was crucial for later calculations involving position-based metrics such as comebacks (difference between grid and finish position) or deficits (loss in position).

IV. Findings and Analysis

We can begin our analysis by looking at our constructors by nationality, and discerning which nationality it is most common for constructors to have. When we look at the data, we can see that most constructors, by an overwhelming majority, are British (***Table 1***), and that out of the British constructors, McLaren, Williams, and Tyrrell are the constructors that have competed in the most races (***Table 2***). Using the results data that we have for each of these constructors, we can make a graph to visualize how these teams have fared over the years.

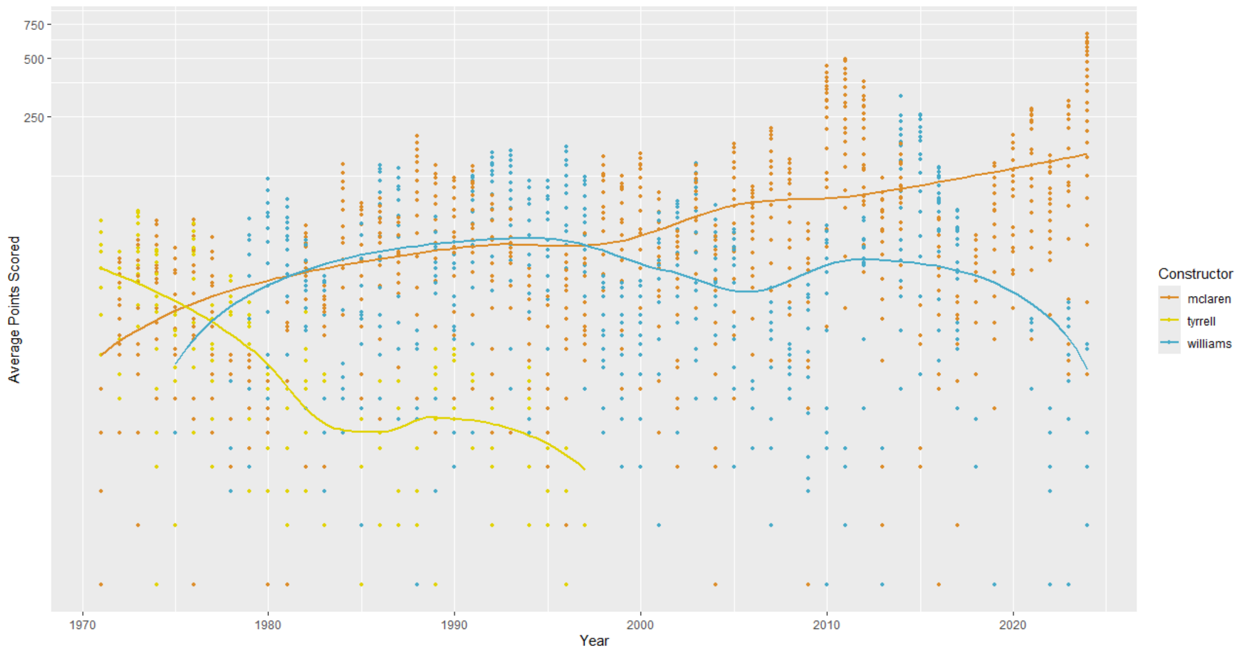

Fig. 1. *Points Comparisons for Top British Constructors*

From this graph, we can see that while Tyrrell started off as the strongest team, they quickly declined and now no longer race on the F1 grid. Williams also enjoyed somewhat of a golden age in the 1980s and 1990s, but McLaren was usually quite close to them, and in recent years, we can see that Williams has sharply declined in performance, reaching the bottom of the championship. McLaren, on the other hand, has been improving in recent years and won the WCC in 2024, showing more promise compared to Williams.

When looking at driver position on the grid, it's quite interesting to see what drivers are the best at making 'comebacks', which we can determine by looking at their final position versus their starting position on the grid. If we don't cap the number of races when trying to determine the best comeback driver, we get lots of drivers that have only raced a few races (***Table 3***). If we institute a minimum floor of races though, say 50, we get drivers that are a bit more seasoned, and we can see that Piercarlo Ghinzani takes the title of best comeback driver (***Table 4***), with an average comeback of 11.2 places (which is over half the grid). If we look at constructors and institute a 200 race limit instead, we see that Osella is the constructor that's the best at comebacks, with an average comeback of 10.8 places (***Table 5***).

If we look at the opposite, say the drivers that lose the most places during a race, we can see that Yuki Tsunoda has the largest average deficit, but only with an average race deficit of 0.338 places (***Table 6***), which over many races is quite negligible. We could consider the fact that a driver with a very large average deficit might be cut from his team before completing 50 races. When we lower the minimum floor to five races, the top deficit driver changes to be Chuck Stevenson (***Table 7***), but changing the minimum doesn't seem to have a large overall effect.

However, it's also worth noting that the ability of a driver to make a comeback may not be reflected if the driver starts in the front of the grid more regularly, where there are less places to make up. If we look at the drivers that start from pole, or first, position the most often, we start to see the names of drivers considered to be the 'greats' of Formula 1 (***Table 8***). This pattern continues when we look at the drivers that are the best at qualifying overall, as most of these drivers are world champions (***Table 9***). Likewise, when we look at the constructors that start from pole most often, we also see the names of classic, established teams, like Ferrari and McLaren (***Table 10***).

Say we want to visualize the performance of Formula 1's top drivers in a geographic sense. First, we can get a general sense of F1's most common circuits (***Table 11***), which the data show as being primarily in Europe. A worldwide and Eurocentric view of the most common F1 circuits are shown on the graphs below, and a graph of circuit altitude is included in the appendix, though circuits do not differ significantly in altitude, save the Mexican GP.

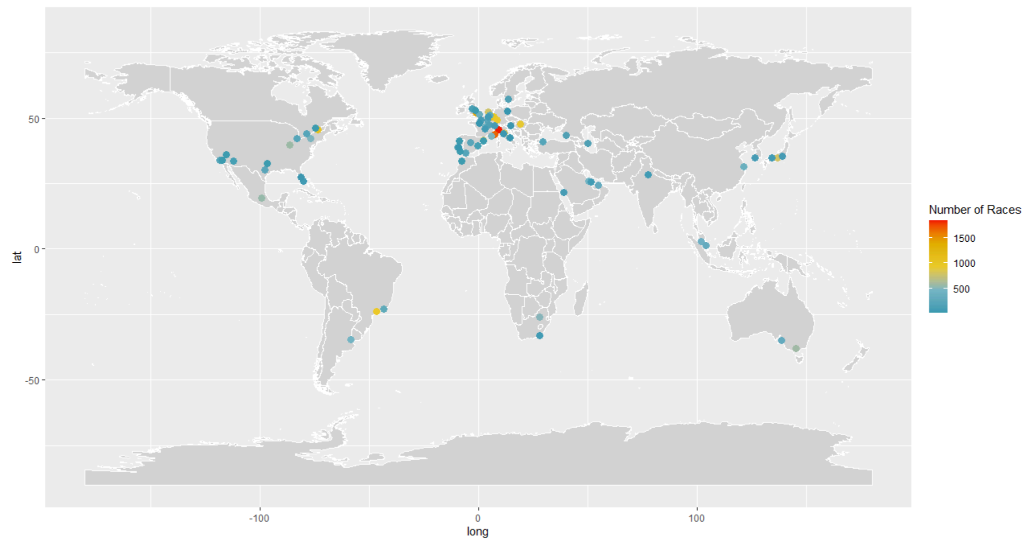

Fig. 2. *Circuit locations and number of races (Worldwide view)*

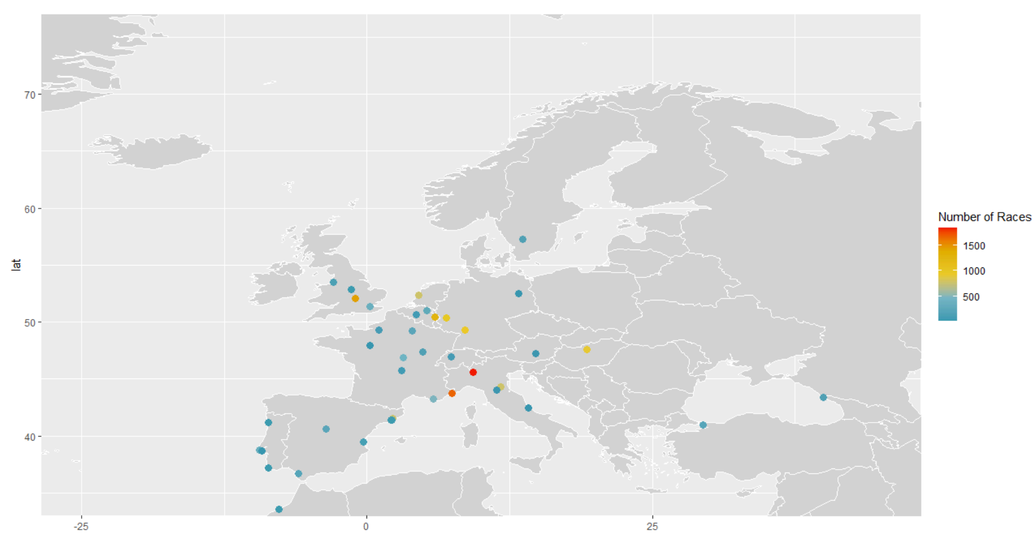

Fig. 3. *Circuit locations and number of races (Eurocentric view)*

Now let's look at the performance of the three most winningest drivers in F1: Lewis Hamilton, Michael Schumacher, and Max Verstappen (***Table 12***). First, we'll take a look at Lewis Hamilton's performance across the world, shown on the graph below.

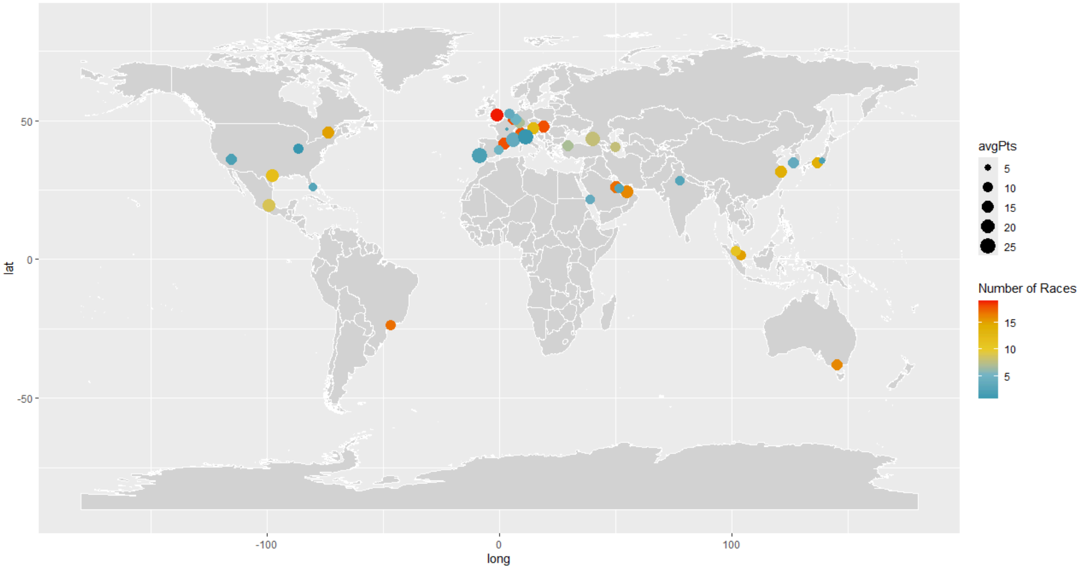

Fig. 5. *Lewis Hamilton's Average Points and Number of Races*

We can see that the circuit Hamilton has raced at the most is Silverstone, or the British GP, which makes sense, seeing that Hamilton is a British racing driver and Silverstone is his home Grand Prix. He also seems to perform especially well in races in Europe and the Middle East, and many of his average points at a given circuit are or exceed 20, which is a further indication of Hamilton's excellent driving caliber.

Now we can look at Michael Schumacher, who raced in an era prior to that of Lewis Hamilton and Max Verstappen.

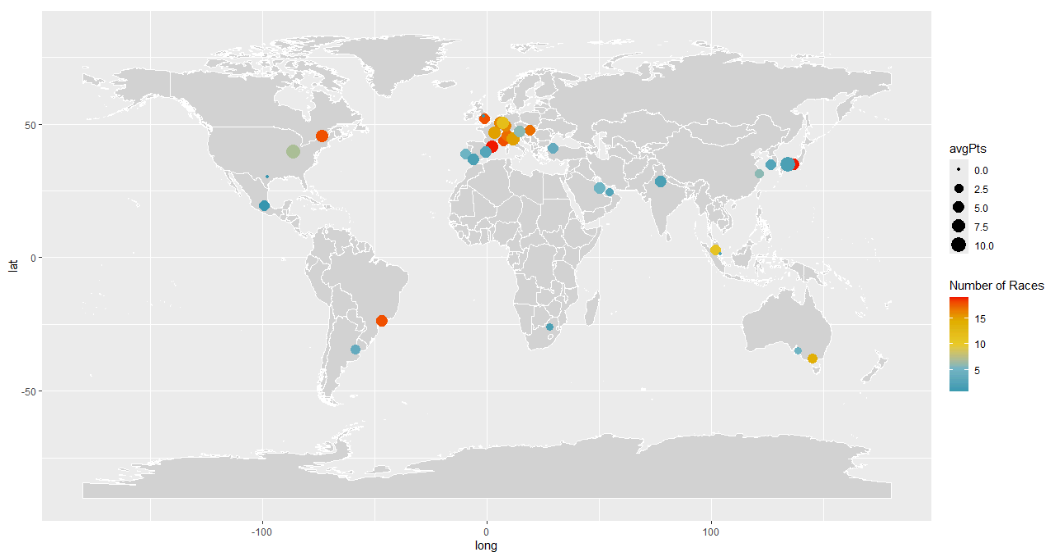

Fig. 6. *Michael Schumacher's Average Points and Number of Races*

His average points are lower, but F1 regulations have changed, and the maximum number of points a driver can score in a race have changed as well. His most raced at circuit is the Barcelona GP, which doesn't have any direct personal connection to Schumacher, but we can see that Schumacher has raced extensively in Europe, and sparsely throughout the rest of the world. This is due to F1 circuit contracts changing over the years, with the sport becoming more international as the years continue. In many ways, Schumacher's graph shows the difference in the F1 era he raced in and the F1 that we know today.

Finally, we can look at our youngest driver in this trio and the current world champion, Max Verstappen.

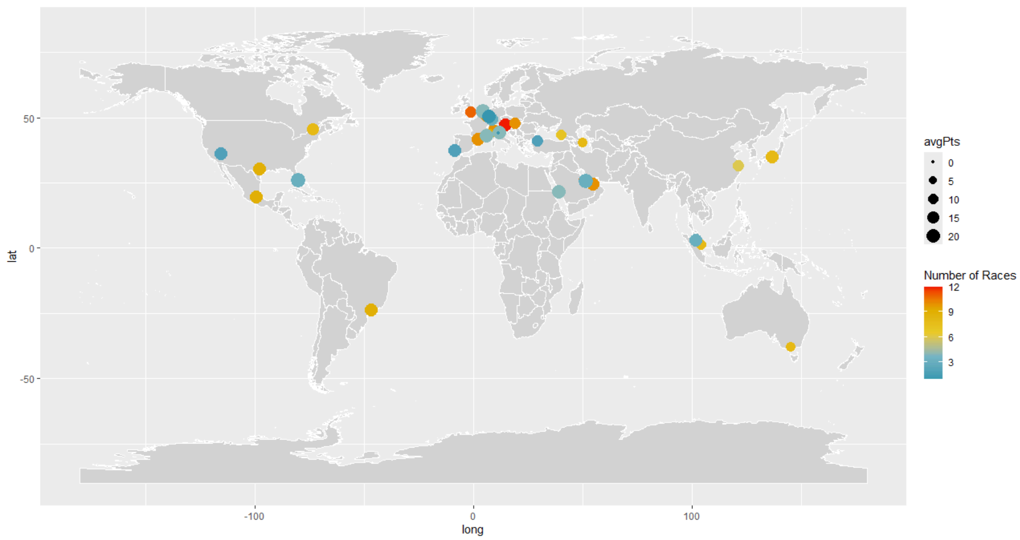

Fig. 7. *Max Verstappen's Average Points and Number of Races*

Verstappen's most raced circuit is the Red Bull Ring, or the Austrian GP, which is the home race of his constructor, Red Bull. While we can see that Verstappen hasn't raced as many years as Hamilton or Schumacher, we can see that his average points at each circuit are excellent, especially for a driver of his age.

Formula 1 can be quite the hereditary sport, with father-son duos being a common occurrence. Let's focus on three father-son pairs: the Rosbergs, the Verstappens, and the Schumachers. All of these pairs have a distinct relationship between the father's performance and the son's performance, which is illustrated by Fig. 8 and Fig. 9, comparing points and position among these pairs.

As we can see, each pair has a unique relationship. For the Rosbergs, both father Keke and son Nico had lots of success in racing, with both becoming world champions. In the case of the Verstappens, father Jos was quite mediocre during his time in F1, while his son Max is a 4-time back-to-back world champion, greatly eclipsing his father's career. The opposite is true for the Schumachers, where father Michael is a 7-time world champion, tied for the most of all time, and son Mick only raced in F1 for two seasons with very little success, mostly keeping to the back of the grid and eventually being replaced.

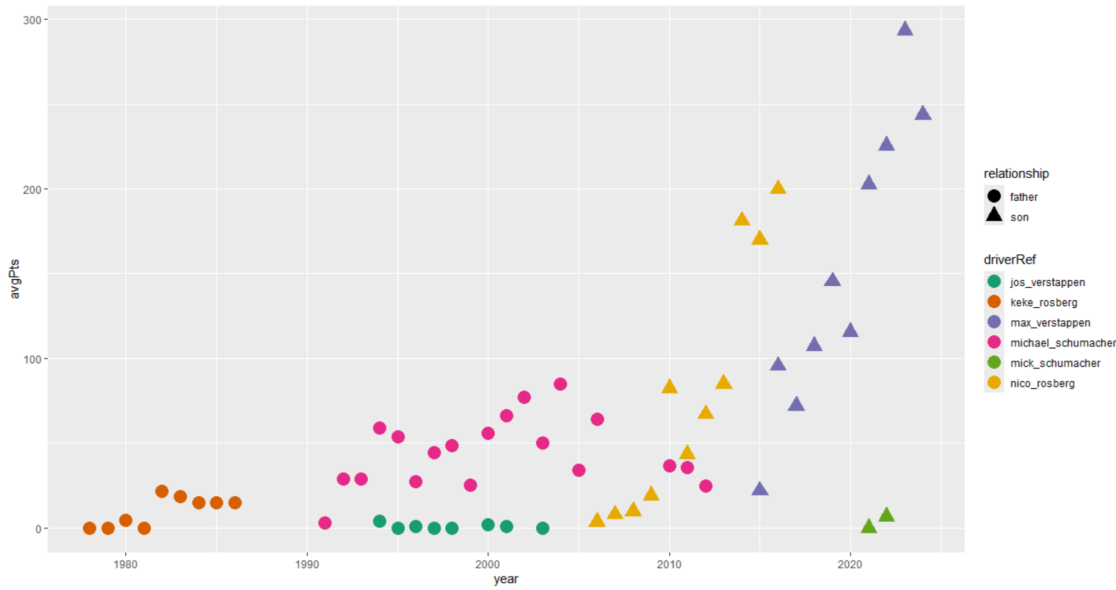

Fig. 8. *Fathers vs. Sons Average Points Comparison*

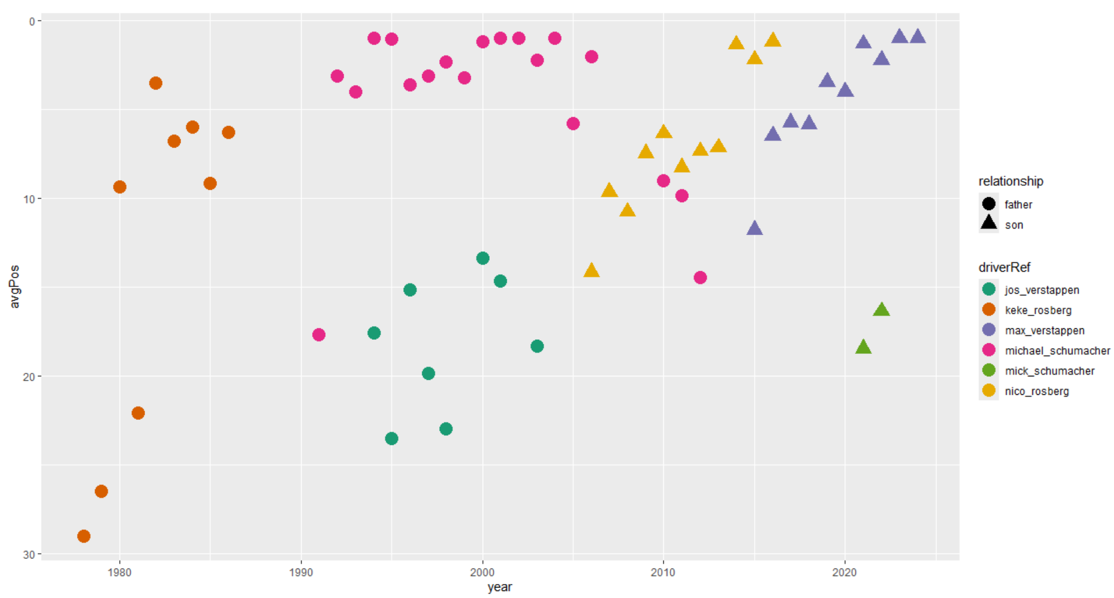

Fig. 9. *Fathers vs. Sons Average Position Comparison*

These differences are further highlighted in the graph below, which indicates a world champion season for a driver with a gold ring. It should be noted, however, that the number of places on the grid has varied over the years. For example, Keke Rosberg being around 29th place in the 1970s translates to Mick Schumacher being around 19th place in the 2020s, since the number of drivers on the grid at a time has changed. This change in seat number is illustrated in Fig. 10 in the appendix.

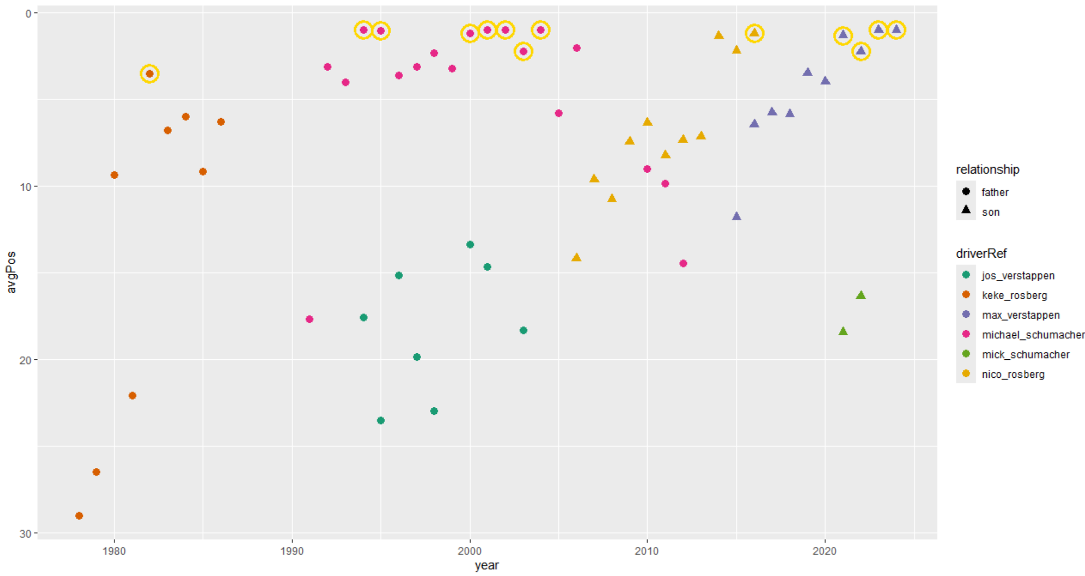

Fig. 11. *Fathers vs. Sons Average Position Comparison (with WC component)*

Finally, we can take a look at world championship years for constructors, and add a driver component in as well. First, we'll limit this analysis to the top eight constructors to maintain graph visibility, that is, the eight constructors with the most races (***Table 17***). Now, when a constructor wins a championship, it does not necessarily mean that one of their drivers also won a championship. The graph below displays this phenomenon, where 'double-win' seasons (both driver and their constructor win a championship) are outlined with a gold ring and single-win seasons (only the constructor wins) are outlined with a blue ring.

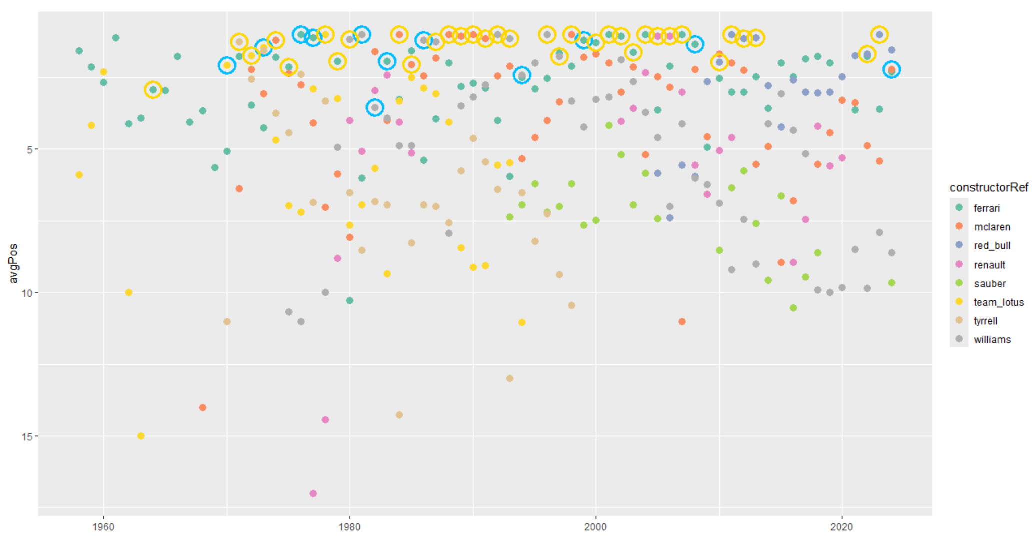

Fig. 12. *Top Eight Constructors Double vs. Single Win Seasons*

As we can see, most constructor world championships are accompanied by a driver championship, but this is not always the case. Additionally, we can see that the top eight teams excludes newer teams that have been performing well, like Mercedes, so we have gaps in our graph from where their championships should be.

Works Cited

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